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CLARA (Clustering Large Applications) (1990)

- CLARA (Kaufmann and Rousseeuw in 1990)
 - Built in statistical analysis packages, such as S+
- It draws multiple samples of **data set**, applies PAM on each sample, gives best clustering as output
- Handle larger data sets than PAM (1,000 objects in 10 clusters)
- Efficiency and effectiveness depends on the sampling

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3.2.4 CLARA

- CLARA (Clustering Large Applications) uses a sampling-based method to deal with large data sets
- A random sample should closely represent the original data
- The chosen medoids will likely be similar to what would have been chosen from the whole data set

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CLARA

- Draw multiple samples of the data set
- Apply PAM to each sample
- Return the best clustering

Choose the best clustering

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CLARA Properties

- Complexity of each iteration is: $O(ks^2 + k(n-k))$
 - s : the size of the sample
 - k : number of clusters
 - n : number of objects
- PAM finds the best k medoids among a given data, and CLARA finds the best k medoids among the selected samples
- Problems
 - The best k medoids may not be selected during the sampling process. In this case, CLARA will never find the best clustering
 - If the sampling is biased we cannot have a good clustering
 - Trade off of efficiency

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CLUSTER ANALYSIS

- Partitioning Methods: **CLARANS** (Clustering Large Applications based upon Randomized Search)
- It starts with an arbitrary set of k medoids
- It picks a medoid and replaces it by a non-medoid. If it is more suitable as a medoid, it is made a medoid
- If it is not a better medoid, a counter is incremented
- If the value of the counter reaches a pre-set number, a local minimum is said to be found and algorithm restarts with a new random set of medoids
- This process is repeated n times and the best set of medoids is retained

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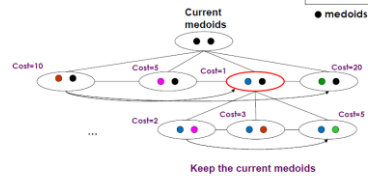
3.2.5 CLARANS

- CLARANS (Clustering Large Applications based upon RANdomized Search) was proposed to improve the quality and the scalability of CLARA
- It combines sampling techniques with PAM
- It does not confine itself to any sample at a given time
- It draws a sample with some randomness in each step of the search

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CLARANS: The idea

Clustering view

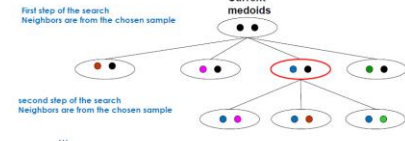


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CLARANS: The idea

CLARA

- Draws a sample of nodes at the beginning of the search
- Neighbors are from the chosen sample
- Restricts the search to a specific area of the original data

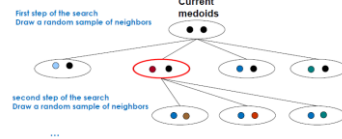


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CLARANS: The idea

CLARANS

- Does not confine the search to a localized area
- Stops the search when a local minimum is found
- Finds several local optimums and output the clustering with the best local optimum



The number of neighbors sampled from the original data is specified by the user

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CLARANS Properties

Advantages

- Experiments show that CLARANS is more effective than both PAM and CLARA
- Handles outliers

Disadvantages

- The computational complexity of CLARANS is $O(n^3)$, where n is the number of objects
- The clustering quality depends on the sampling method

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Summary of Section 3.2

- Partitioning methods find sphere-shaped clusters
- K-mean is efficient for large data sets but sensitive to outliers
- PAM uses centers of the clusters instead of means
- CLARA and CLARANS are used for clustering large databases

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